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Multi-scale modeling of human colonic crypt fission using a level set method

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1 Introduction

Extensive biological work and mathematical modeling has been done in investigating the effects of the canonical Wnt Pathway in the human colonic crypt [1–7]. This basic, yet complex, network of proteins is the key to normal cell function. At the same time, abnormal regulation of this pathway leads to developmental defects as well as cancer initiation [8–10]. The Wnt pathway is responsible for the regulation of cytoplasmic β-catenin, which is linked to cellular proliferation and differentiation.

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Wnt induced β-catenin activity is also regulated by the protein labeled APC, which is encoded by the tumor suppressor gene also labeled *APC*. This protein has been linked to the initiation of colon cancer for some time [11–14]. Mathematical modeling of the regulation and mutation of *APC* has shed light on how this pathway influences and results in cancerous states within the human colon [15–19]. These models continue to be used in larger, multi-scale models and will be used in the model presented here.

The biochemical profiles of the protein concentrations involved in the Wnt pathway play an especially important role within the phenotypical layout of the human colonic crypt [8,20–22]. The fingerlike invagination known as the crypt is responsible for continuous renewal of the epithelium lining of the large intestine. Cell lineages that line the crypt walls are constantly splitting and differentiating until they finally reach the top of the crypt where they are extruded into the intestinal lumen [23,24]. Evidence shows that continuous stimulation of Wnt activity in the base of the crypt is paramount for stem cell maintenance and crypt homeostasis [21,22]. Also, the amount of intracellular APC increases almost fivefold as measured from the bottom of the crypt to the top [25,26]. Inverse gradients of APC and Wnt induced β-catenin are responsible for the zone of rapid proliferation approximately one-third of the way up the crypt. Most of the cell movement and crypt formation is attributed to this position-dependent zone of rapid cell production [27]. Initiation of sporadic colon carcinomas is characterized by an accumulation of cells within the crypt. This is especially evident in most hereditary cases of familial adenomatous polyposis (FAP), where *APC* gene mutations create a diminished APC gradient that mimics the effect of the Wnt pathway further up the the crypt thereby creating an accumulation of β-catenin within the epithelial cells [28–30]. The imbalance of key Wnt pathway proteins brought on by successive *APC* mutations results in a larger proportion of proliferative cells, ultimately resulting in fully proliferative crypts and a complete loss in crypt structure.

The development of polyps and the progression of adenoma morphogenesis is accelerated by abnormal crypt fission [27,31]. The normal crypt cycle is a mechanism by which the crypt itself divides to form two identical crypts [32–36]. This process is characterized by an initial growth of the crypt followed by a budding/indentation at the base. The crypt bifurcation elongates and extends upwards until it reaches the colonic lumen, finishing the crypt fission phase. The crypt cycle is a slow but continuous process that is responsible for epithelial maintenance and essential for injury repair. Totafurno et. al. and Bjerkness et. al. suggest that crypt size and the stem cell population govern crypt fission [35,37]. Boman et. al. suggest that crypt fission is governed by the counter-current Wnt-APC gradients [27]. In this way, a zone of rapid proliferation, termed the "sweet spot," is established in the base of the crypt that is responsible for initiating the budding process and eventually forming the new crypt. Figure 1 shows the progression of the crypt fission process based on the inverse gradient hypothesis.

In the presence of APC mutations, tissue disorganization is a result of an increase in the rate of crypt fission [38–41]. Wasan et. al. demonstrate an increased rate of crypt fission in FAP patients and Bjerknes et. al. show that the rate is even faster in homozygote mutated crypts [31,37]. Furthermore, these studies show that the crypt budding and fission process is asymmetrical, leading to branching and non-identical

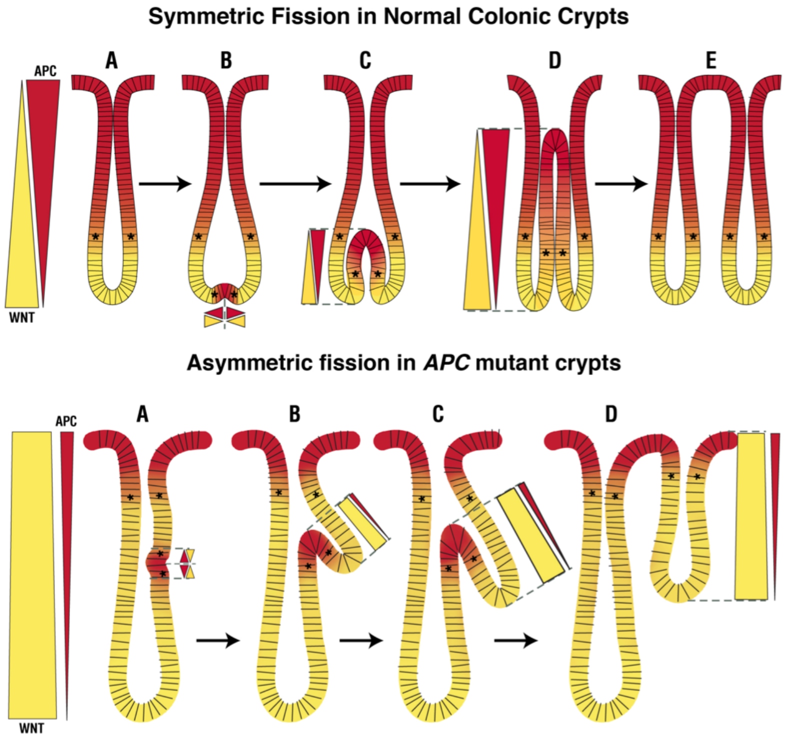

Fig. 1: Schematic that illustrates both normal and abnormal crypt fission. (Top) Symmetric crypt fission that initiates at the base of the crypt resulting in a normal crypt cycle, which involves (A) growth, (B) budding, (A→ B) bifurcation, (B→E) fission phases. (Bottom) Asymmetric crypt fission characterizes the abnormal crypt cycle, which results in two crypts of different sizes. Abnormal crypt fission occurs through a budding/bifurcation phase that starts somewhere along the crypt axis. The star, which denotes the "sweet spot," is an indicator of the zone of rapid proliferation that ultimately determines the bifurcation point.

crypts. It is apparent that APC plays a role in normal and abnormal crypt fission. Boman et. al. suggest the proliferative zone shift due to a diminished APC gradient is responsible for atypical budding and branching [27]. In normal crypts, the proliferative region is kept near the base; thereby, restricting the initiation of crypt fission at the lower region of the crypt. In this scenario, crypt fission will commence in a symmetric manner yielding two crypts of the same size. If the APC gradient is diminished, the proliferative zone shifts upward creating a larger region where budding can occur. In this case, budding can start on the side of the crypt column (see Figure 1), which eventually leads to branching [27]. Excessive branching underlies the tissue disorganization found in adenomas.

Over the past decade, considerable attention has been given to modeling the dynamics within a static crypt. The most extensive individual-based model was presented by Leeuwen et. al. in 2010. This was a multi-scale model that used a Delaunay triangulation to identify cell locations. Cell division is based on intracellular levels of Wnt induced proteins [16]. Cell motion is derived from another model presented by Meineke et. al., who assume that each neighboring cell within the crypt is connected by a linear, over-damped spring [15]. In this sense, motility of cells is created by mitotic pressure, i.e. crowding via cell mitosis forces the cells to fill the limited space and thus move along the crypt wall. In this model, Wnt target proteins at the intracellular level act as a driver for cell proliferation. Regulation of the Wnt pathway is governed by the Lee-Heinrich model [1]. A coarse-grained continuous model was formulated by Murray et. al. in 2011, which tracks a concentration of cells along the major crypt axis as it moves away from the Wnt signal [42]. Murray et. al. were able to effectively compare the results of both the discrete and continuous model, showing a strong correlation between the two models. Emerick et. al. adopted this continuous strategy and defined the concentration of cells as $p = p(s, \beta, \tau)$, which is a function of crypt level, s, β-catenin concentration (measured as a spatial dimension), β, and time, τ. Following the work of Murray et. al., they applied a conservation of cell number to obtain a PDE of the form

$$\frac{\partial \hat{p}}{\partial \hat{\tau}} + \frac{\partial J_s}{\partial \hat{s}} + \frac{\partial J_\beta}{\partial \hat{\beta}} = r, \tag{1}$$

where J_s and J_β denote the flux of cells in $\hat{s}$ and $\hat{\beta}$, respectively, and r represents a net production of cells due to cell mitosis. The flux of cells in the $\hat{s}$ direction, J_s, is assumed to be convective in nature. The flux of cells in the β-catenin regime, J_β, is both convective and diffusive and governed by the APC regulation model at the intracellular level, which was presented by Emerick et. al. in 2016 [7]. Using this PDE model of the static crypt, Emerick et. al. were able to show that when the model is subjected to a diminished APC gradient, as is the case when mutated *APC* is present, a larger proportion of proliferating cells is present within the crypt. However, this model uses a static crypt domain, and hence the effects of crypt fission are not considered.

Less attention has been focused on modeling the dynamics of crypt fission; however, the future of colonic crypt modeling is in that direction. Figueiredo et. al. recently created a model to describe aberrant crypt morphogenesis [43]. In this model, cell populations are assumed to move in the crypt as a fluid moves through a porous media, obeying Darcy's law [44–46]. In this sense, a system of transport/mass conservation equations are formed to describe the space-time evolution of a population of proliferative and apoptotic cells. Furthermore, the cell movement equations are coupled to the crypt geometry, which is modeled by a two dimensional "U" shape. Crypt budding occurs through a build up of cells at the base of the crypt, which applies pressure to the crypt boundary. The authors present several simulations in the normal and abnormal cases and compare the results to medical imaging.

Another interesting crypt fission model couples the activity of the Wnt and BMP pathways to proliferative and differentiated cell populations [25]. The concentration of the pathway components are modeled through a reaction-diffusion system along the major crypt axis. The space-time evolution of the crypt boundary are modeled using an energy functional in two-dimensional space that is dependent on cell concentrations. The energy functional is created in such a way that the location of proliferating cells yields the minimum energy, thereby creating the desired spatial cell concentration profiles along the crypt wall. This also generates a curvy crypt shape, whose growth and division is driven by the underlying Turing pattern of the reaction-diffusion system.

The model of crypt fission discussed in this paper is an extension of the static crypt model presented by Emerick et. al. [19]. In this paper, we track only the cell concentration along the major crypt axis by eliminating the β-catenin variable, i.e.

$$\hat{q}(\hat{s},\hat{\tau}) = \int_0^{\beta_{max}} \hat{p}(\hat{s},\hat{\beta},\hat{\tau})\, d\hat{\beta},$$

where β_{max} is a necessary upper bound on the amount of β-catenin that can be found in the typical epithelium cell. As presented in Emerick et. al., the dimensional PDE model in cell number density, $q(s,\tau)$, is shown for convenience below:

$$\begin{cases} \dfrac{\partial \hat{q}}{\partial \hat{\tau}} = \dfrac{\partial}{\partial \hat{s}}\left(\dfrac{\alpha}{\hat{q}^2}\dfrac{\partial \hat{q}}{\partial \hat{s}}\right) + H(\hat{s}_p - \hat{s})\dfrac{\ln 2}{T_c}\hat{q} & \text{for } (\hat{s},\hat{\tau}) \in (0,\hat{L}) \times (0,\hat{T}] \quad (2) \\ \dfrac{\partial \hat{q}}{\partial \hat{s}}(0,\hat{\tau}) = 0 & \text{for } \hat{\tau} \in (0,\hat{T}] \quad (3) \\ \hat{q}(\hat{L},\hat{\tau}) = 1/\ell & \text{for } \hat{\tau} \in (0,\hat{T}] \quad (4) \\ \hat{q}(\hat{s},0) = \hat{q}_0(\hat{s}) & \text{for } \hat{s} \in (0,\hat{L}). \quad (5) \end{cases}$$

We note that the non-linear diffusion presented here is originally derived from a coarse-grained model that considers cells being connected via over-damped linear springs [15,16]. Here, α is the ratio of the spring constant to the cell viscosity, ℓ is the resting spring length (typically set to 1), T_c is the average doubling time of a cell in the proliferative region, and the value s_p is a threshold value along the crypt that determines the proliferative zone based on the intracellular level of stemness. We investigate the normal crypt growth and fissioning process by implementing a moving boundary into the model considered by Emerick et. al. in [19] using Equations (2) – (5) as a starting point. We seek to model crypt formation dynamics and understand how the underlying processes of APC regulation might affect crypt fission. We present an extension to the one-dimensional model considered by Emerick et. al. that includes a mechanism to initiate crypt fission from cell crowding in the base. However, the assumption that the crypt is radially symmetric fails during crypt fission since symmetry is broken during the division process. In the last part of this paper, we consider a geometric method that describes normal crypt fission in three-dimensional space. We then couple this dividing process to the PDE model by way of the arc-length of each crypt level. Our goal is to develop a framework for modeling the normal crypt fissioning process and to couple this process to the underlying APC regulation model.

This paper is organized as follows. In Section 2, we implement a moving boundary element via the single phase Stefan condition into the reduced PDE model of Emerick et. al. [19]. We discuss the model dynamics and fit new parameters to normal crypt data. In Section 3, we formulate a level set method to describe the normal crypt fissioning process in three-dimensions. In Section 4, we couple the protein flux PDE model presented by Emerick et. al. [19] to the level set crypt fissioning process.

This provides a framework for determining the effects of APC regulation on crypt fission. We end the chapter with a discussion in Section 5.

2 Stefan Condition

We create a model that can describe the formation of a new crypt starting with only a single cell at the base of an already formed crypt. To this end, we employ a Stefan condition at the boundary $s = L$ on the Emerick et. al. model [19] so that the proliferative region pushes the last cell to create a longer, one-dimensional domain. In this sense, when crypt fission is initiated via cell crowding at the base, a new chain pushes its way up to create a new crypt wall. The Stefan problem normally considers a moving boundary or phase change as applied to the melting of ice [47], where the advancing liquid/solid interface is coupled to the change in temperature at the boundary. We implement a similar condition to the system given by Equations (2) – (5), without the cell production term, to first illustrate how the mechanism behaves. Then, we introduce cell proliferation and cell death to create the normal crypt domain.

In the following analysis, we denote dimensional variables using hat notation. We consider a cell number density, $\hat{q}(\hat{s},\hat{\tau})$, whose evolution is governed by the system in Equations (2) – (5) with no cell proliferation. Assume $\hat{s} \in [0, \hat{L}(\tau)]$, where the change in $\hat{L}$ is governed by the balance equation known as the Stefan condition

$$\frac{d\hat{L}}{d\hat{\tau}} = -\frac{\alpha}{\hat{q}^3}\frac{\partial \hat{q}}{\partial \hat{s}}\bigg|_{\hat{s}=\hat{L}(\tau)}, \tag{6}$$

which holds for all $\hat{\tau} > 0$. Here, we note that there is no constant of proportionality as the speed of the growing length is exactly that of the convective velocity of the last cell, which is given by the right hand side of Equation (6). We also implement the previous boundary conditions on $\hat{q}$ so that there is always one cell at the moving boundary and it is in the equilibrium state. Therefore, our system becomes

$$\begin{cases} \dfrac{\partial \hat{q}}{\partial \hat{\tau}} = \dfrac{\partial}{\partial \hat{s}}\left(\dfrac{\alpha}{\hat{q}^2}\dfrac{\partial \hat{q}}{\partial \hat{s}}\right) & \text{for } (\hat{s},\hat{\tau}) \in \big(0,\hat{L}(\hat{\tau})\big) \times (0,\hat{T}] & (7) \\ \dfrac{d\hat{L}}{d\hat{\tau}} = -\dfrac{\alpha}{\hat{q}(\hat{L},\hat{\tau})^3}\dfrac{\partial \hat{q}}{\partial \hat{s}}(\hat{L},\hat{\tau}) & \text{for } \hat{\tau} \in (0,\hat{T}] & (8) \\ \dfrac{\partial \hat{q}}{\partial \hat{s}}(0,\hat{\tau}) = 0 & \text{for } \hat{\tau} \in (0,\hat{T}] & (9) \\ \hat{q}(\hat{L},\hat{\tau}) = 1/\ell & \text{for } \hat{\tau} \in (0,\hat{T}] & (10) \\ \hat{q}(\hat{s},0) = q_0(\hat{s}) & \text{for } \hat{s} \in \big(0,\hat{L}(\hat{\tau})\big). & (11) \\ \hat{L}(0) = L_0 & & (12) \end{cases}$$

We can gain some insight into the dynamics of the model by computationally solving the system above using the boundary immobilization method. This method essentially uses the following change of variable in the spatial dimension,

$$s = \frac{\hat{s}}{\hat{L}(\hat{\tau})}.$$

Using this scaling, the problem is cast into the stationary domain $s \in [0,1]$, with new coordinates $(s,\hat{\tau})$. From this, our operators become

$$\frac{\partial}{\partial \hat{s}} = \frac{1}{\hat{L}(\hat{\tau})}\frac{\partial}{\partial s}, \qquad \left.\frac{\partial \hat{q}}{\partial \hat{\tau}}\right|_{\hat{s}} = \frac{\partial \hat{q}}{\partial s}\frac{\partial s}{\partial \hat{\tau}} + \left.\frac{\partial \hat{q}}{\partial \hat{\tau}}\right|_{s} = -\frac{s}{\hat{L}(\hat{\tau})}\frac{d\hat{L}}{d\hat{\tau}}\frac{\partial \hat{q}}{\partial s} + \left.\frac{\partial \hat{q}}{\partial \hat{\tau}}\right|_{s}.$$

Applying this to our PDE, and changing the boundary conditions gives the following dimensional system (where only $\hat{s}$ has been non-dimensionalized)

$$\begin{cases} \dfrac{\partial \hat{q}}{\partial \hat{\tau}} = \dfrac{s}{\hat{L}}\dfrac{d\hat{L}}{d\hat{\tau}}\dfrac{\partial \hat{q}}{\partial s} + \dfrac{\alpha}{\hat{L}^2}\dfrac{\partial}{\partial s}\left(\dfrac{1}{\hat{q}^2}\dfrac{\partial \hat{q}}{\partial s}\right) & \text{for } (s,\hat{\tau}) \in (0,1)\times(0,\hat{T}] \quad (13) \\ \dfrac{d\hat{L}}{d\hat{\tau}} = -\dfrac{\alpha}{\hat{L}\hat{q}(1,\hat{\tau})^3}\dfrac{\partial \hat{q}}{\partial s}(1,\hat{\tau}) & \text{for } \hat{\tau} \in (0,\hat{T}] \quad (14) \\ \dfrac{\partial \hat{q}}{\partial s}(0,\hat{\tau}) = 0 & \text{for } \hat{\tau} \in (0,\hat{T}] \quad (15) \\ \hat{q}(1,\hat{\tau}) = 1/\ell & \text{for } \hat{\tau} \in (0,\hat{T}] \quad (16) \\ \hat{q}(s,0) = \hat{q}_0(s) & \text{for } s \in (0,1). \quad (17) \\ \hat{L}(0) = \hat{L}_0 & \quad (18) \end{cases}$$

We note here that $\partial \hat{q}/\partial s \leq 0$ for $s \in [0,1]$ and $\hat{\tau} > 0$, which means the first term on the right hand side is negative since $d\hat{L}/d\hat{\tau} > 0$. Therefore, this term acts as a sink on the stationary grid which dilutes the cell level density as the length grows. Knowing the dynamics of the previous model without domain growth and noting that there is no source of additional cells, we can assume this system reaches a steady-state. Also, as $L \to \infty$, we can see that both the dilution term and the non-linear cell motion term tend to zero. Hence, we expect the cell number density to approach the spatially homogeneous equilibrium value of $\hat{q}(s,\hat{\tau}) = 1$ as $\hat{\tau} \to \infty$. This is confirmed by solving the steady-state form of the equation.

To discretize the right hand side of Equation (13), we use second order finite differences on a fine grid. We use MATLAB's `ode15s` to efficiently discretize the time variable and numerically integrate the system. We consider a simulation with a $\hat{L}_0 = 4$ and $\hat{q}_0 = 1 + 2H(2-\hat{s})$. In this case, we initially have a group of six cells in the lower half of the domain, and two cells at equilibrium in the upper half. We expect the cells to equilibrate to $\hat{q}(s,\hat{\tau}) = 1$ over time, which will yield a total length of $\hat{L} = 8$. Figure 2 shows the evolution of $\hat{q}(\hat{s},\hat{\tau})$ from its initial configuration to steady-state. We can see the buildup of cells in the base creates a force that advances the last cell until all cells in the crypt are exactly one cell length apart. We also note the total amount of cells are conserved as no cells are lost due to cell death, but rather just displaced. In this case, there are always eight cells in the crypt, but the length of the crypt grows.

The number of cells in the crypt remains constant if there is no source term. If we consider a constant in-flow of cells in the base of the crypt as before, then the length of the crypt will grow indefinitely and the base of the crypt will experience exponential cell growth. Indeed, if we consider a cell proliferation term determined

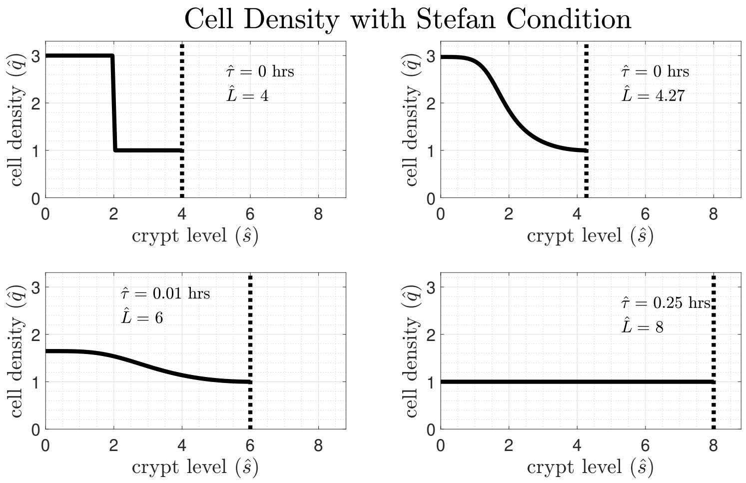

Fig. 2: The evolution of the cell number density, $\hat{q}$, using cell motion as described by the non-linear diffusion coefficient, $D(\hat{q}) = \alpha/\hat{q}^2$, with the Stefan condition. Time $\hat{\tau}$ is in hours and $\alpha = 1840.3$.

by the intracellular level of stemness as in Emerick et. al. [19] and apply the following scalings

$$q = \ell\hat{q}, \qquad \tau = \frac{\ln 2}{T_c}\hat{\tau}, \qquad s = \frac{\hat{s}}{L}, \qquad L = \frac{\hat{L}}{L^*},$$

where L^* is the characteristic length of the crypt, then we obtain the non-dimensional analog of Equation (2)

$$\frac{\partial q}{\partial \tau} = \frac{s}{L}\frac{dL}{d\tau}\frac{\partial q}{\partial s} + \frac{1}{\gamma}\frac{\partial}{\partial s}\left(\frac{1}{q^2}\frac{\partial q}{\partial s}\right) + H(s_p - s)q,$$

which includes the dilution term due to the Stefan condition, which, in non-dimensional terms, is given as

$$\frac{dL}{d\tau} = \underbrace{-\left[\frac{1}{\gamma q(1,\tau)^3}\frac{\partial q}{\partial s}(1,\tau)\right]}_{v_s(1,\tau)>0} L,$$

where v_s denotes the non-dimensional cell velocity. Here, we have

$$\gamma := \gamma\big(L(\tau)\big) = \frac{L(\tau)^2 \ln 2}{(L^*)^2 \ell^2 \alpha T_c}.$$

As noted in Emerick et. al. [19], there exists a threshold value, γ^*, that defines the balance between cell motion and cell proliferation. If $\gamma > \gamma^*$, an exponential increase of cells occurs in the base of the crypt since these cells are moving up the crypt too slowly relative to the speed at which they are being created. In the moving boundary problem, we note that γ is a function of L, which is increasing with time. We also note the the velocity of the last cell is always positive because cell production in the base will continue to advance the last cell. That is, as $\tau \to \infty$, we have $L \to \infty$ since $dL/d\tau = v_s(1,\tau)L$ and $v_s > 0$ for all $s \in [0,1]$. Therefore, as $\tau \to \infty$, we must have $\gamma \to \infty$, which drives cell motion to zero, while the cell proliferation term remains unscaled by L. Further, $dL/d\tau \propto 1/L$, suggesting that L grows as $\mathscr{O}(\sqrt{\tau})$. Thus, the γ^* threshold is ultimately crossed causing an exponential increase of cells in the base of the infinitely growing crypt.

During normal crypt formation and fission, the length of the crypt eventually stops growing as cells at the top are shed into the intestinal lumen. That is, once a balance of cell movement and shedding is reached, the length of the crypt remains constant and cell proliferation and motion regulate normal colonic processes. Therefore, it is necessary to implement a cell shedding zone at the top of the crypt. As cells migrate up the crypt, they proliferate and differentiate, ultimately becoming a terminally differentiated cell by the time they are extruded through the top of the crypt. The level of intracellular stemness acts to identify cells that are both proliferative and terminally differentiated. In this sense, when a cell reaches a minimum level of stemness, it is considered terminally differentiated and will be shed from the crypt.

To incorporate this phenomenon into our current model, we define a threshold stemness value, σ_s, that represents the maximum stemness required for cell shedding. Typically, shedding occurs where the top of the crypt meets with the intestinal lining. Therefore, we prescribe the threshold at the point along the crypt where the wall is slightly rounded to meet the lining. This occurs at approximately crypt level $s = 77$ in our parameterization. The corresponding normal stemness value using the APC regulation model of Emerick et. al. in [7] at this level is approximately $\sigma_s = 0.197$. Using the Heaviside function, we incorporate the following source and sink into the system given by Equations (7) – (12),

$$\left[\frac{\ln 2}{T_c} H(\hat{s}_p - \hat{s}) - mH(\hat{s} - \hat{s}_s)\right] q,$$

where the source is the same as before and m denotes the average cell shedding rate. Here, $\hat{s}_p$ and $\hat{s}_s$ are the cell levels along the crypt that correspond to proliferating stemness, σ_p, and shedding stemness, σ_s, respectively. Using the same scalings as above, we arrive at the following non-dimensional system.

$$\begin{cases} \dfrac{\partial q}{\partial \tau} = \dfrac{s}{L}\dfrac{dL}{d\tau}\dfrac{\partial q}{\partial s} + \dfrac{1}{\gamma}\dfrac{\partial}{\partial s}\left(\dfrac{1}{q^2}\dfrac{\partial q}{\partial s}\right) & \\ \qquad + \left[H(s_p - s) - \mu H(s - s_s)\right] q & \text{for } (s,\tau) \in (0,1)\times(0,T] \quad (19) \\ \dfrac{dL}{d\tau} = -\left[\dfrac{1}{\gamma q(1,\tau)^3}\dfrac{\partial q}{\partial s}(1,\tau)\right] L & \text{for } \tau \in (0,T] \quad (20) \\ \dfrac{\partial q}{\partial s}(0,\tau) = 0 & \text{for } \tau \in (0,T] \quad (21) \\ q(1,\tau) = 1 & \text{for } \tau \in (0,T] \quad (22) \\ q(s,0) = q_0(s) & \text{for } s \in (0,1). \quad (23) \\ L(0) = \dfrac{L_0}{L^*} & \quad (24) \end{cases}$$

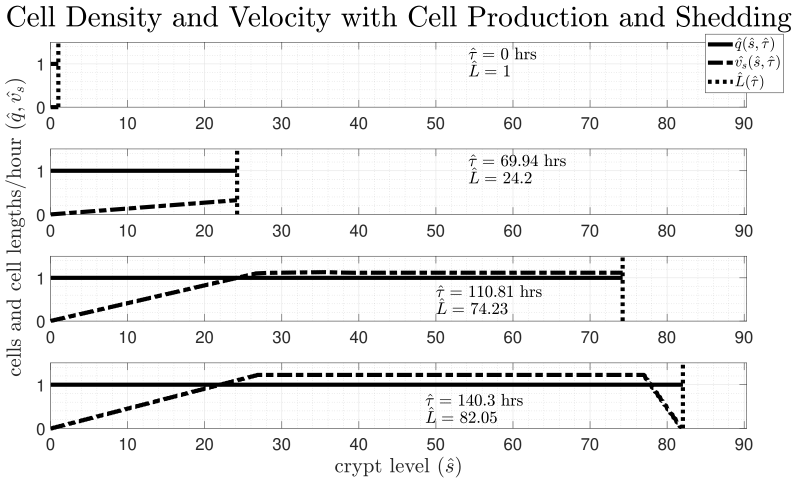

Fig. 3: A simulation of Equations (19) – (24) that represents normal crypt growth. (From top to bottom) The evolution of the cell number density, $\hat{q}$, and cell velocity, $\hat{v_s}(\hat{s},\hat{\tau})$, using cell motion as described by the non-linear diffusion coefficient, $D(\hat{q}) = \alpha/\hat{q}^2$ with the Stefan condition. Parameters: $\alpha = 1840.3$, $\mu = 5.428$, $\sigma_p = 10.42$, and $\sigma_s = 0.197$.

Here, $\mu = T_c m/\ln 2$ measures the ratio of the cell proliferation time scale to the cell shedding time scale. Since γ, s_p, and s_s are fixed to normal crypt data according to previous methods (see [19]), the parameter μ uniquely determines the final length of the crypt. This is assuming that it takes approximately 5 days to form a new crypt, which is the same as the crypt renewal time. Here, we will assume that it takes approximately the same time, but quantitative data for crypt formation and fission time is not well known. To obtain a feasible value for μ, we consider the normal length of the crypt, which is $L^* = 82$ cell diameters. From here, we can select a normal shedding rate as $\mu = 5.428$, which means the cells are shedding out of the top nearly five times faster than they are created in the base.

Figure 3, shows the evolution of the cell number density, cell velocity, and crypt length as time progresses. The figure presents the results in dimensional units. We can see that the crypt is growing as cells are being produced in the proliferative zone. The cell velocity is similar as before since the cells steadily move faster until the edge of the proliferative zone is reached. The cell velocity and crypt length steadily increase until cells with a stemness value below σ_s are present. Once this occurs, cells begin to shed into the intestinal lumen and we see a drop in the cell velocity. This is due to less cells being bunched together at the top. They are closer to equilibrium and hence tend to move slower. This is necessary for the length of the crypt to stop growing since $dL/d\tau$ is balanced by the cell velocity. Even with this slight change in velocity, the crypt is formed in 4.80 days, which can be seen from the time-course plot in Figure 3. Furthermore, at steady-state there are approximately 3334 cells and about 991 of them are proliferative cells, which reflects the normal crypt dynamics.

This model establishes a method by which a new crypt is formed from a single cell. Because we are still assuming the crypt is radially symmetric, the model creates the crypt one cell ring at a time. This simple crypt formation model, therefore, does well to simulate the creation of a crypt from a single cell. However, as we'll see, to extend this model to include crypt fission, we will have to account for the break in radial symmetry as the crypt divides in three-dimensional space.

3 Level Set Method

We implement a method to describe the normal crypt fissioning process in three dimensional space. As seen in the last section, as the crypt bifurcates, radial symmetry is broken, and so the one-dimensional model cannot be used to accurately simulate crypt fission. Normal crypt fission begins at the base, with a dividing plane that bifurcates the crypt upward. To replicate this process, we consider the crypt as a stack of cell rings such that each ring divides into two identical rings in a cascade-type fashion that begins with one cell at the base. In this sense, the crypt division process is captured geometrically and we couple this dynamic to the Stefan problem considered in the last section.

To this end, we consider the level set method to geometrically describe the changing domain at each crypt level. Since the length of the domain at each crypt level contributes to the total number of cells, this will allow us to monitor the change in cell density as the crypt divides. We consider a coupled set of partial differential equations to describe the global dividing process.

The level set method is an initial value problem formulation that is used to describe interface propagation. Osher and Sethian produced the pioneering work in this field [48,49]. For our purposes, we adapt their method to the crypt fission problem by considering a moving two-dimensional front in (ξ,ζ) space. If we suppose that the curve moves in space with some speed F, potentially positive or negative at any given time, then the front can move either forward or backward over a point several times. To track the front, we link the two-dimensional curve to a level set of a higher dimensional function, ϕ. An evolutionary PDE in ϕ can lend a way to track the propagation of the front if we let the curve be equal to the zero level set of the function ϕ at any given time. That is, let $\boldsymbol{\xi}(\tau) = \big(\xi(\tau),\zeta(\tau)\big)$ be the time dependent parameterization of the curve, then

$$\phi\big(\boldsymbol{\xi}(\tau),\tau\big) = 0$$

for all $\tau \geq 0$. Differentiating with respect to τ and using the chain rule gives

$$\frac{\partial \phi}{\partial \tau} + \nabla\phi\big(\boldsymbol{\xi}(\tau),\tau\big)\cdot\boldsymbol{\xi}'(\tau) = 0.$$

Here, we assume the speed F is prescribed in the normal direction of the moving curve so that $\boldsymbol{\xi}'(\tau)\cdot\boldsymbol{n} = F$, where $\boldsymbol{n}$ is the unit normal vector given by $\boldsymbol{n} = \nabla\phi/|\nabla\phi|$. Hence, we obtain the equation

$$\frac{\partial \phi}{\partial \tau} + F|\nabla\phi| = 0. \tag{25}$$

We assume we begin with an initial function, $\phi\left(\boldsymbol{\xi},0\right)$, that gives our initial curve $\Gamma_{\text{init}}(\boldsymbol{\xi}) = \left\{\boldsymbol{\xi} \,\middle|\, \phi(\boldsymbol{\xi},0) = 0\right\}$. This is the level set formulation given by Osher and Sethian [49] and it describes the motion of the interface for any speed function F.

In our circumstances, we want to apply Equation (25) to each level of the crypt. We consider a stack of initial curves given by

$$\Gamma_{\text{init}}(\boldsymbol{\xi};s) = \left\{\boldsymbol{\xi} \,\middle|\, |\boldsymbol{\xi}| = x(s)\right\},$$

where $x(s)$ is the radius of the crypt at level s given by the normal crypt parameterization. Therefore, for each s, we have a circle of radius $x(s)$ and we wish to propagate this circle into two circles of equal size using the level set method. To this end, we consider the level set function that corresponds to our initial set of circles. Here, we are referring to a cone (for each s) of the form

$$\phi(\boldsymbol{\xi},0;s) = \max\left\{0, x(s) - |\boldsymbol{\xi}|\right\},$$

such that setting $\phi(\boldsymbol{\xi},0;s) = 0$ gives Γ_{init}. Now, we prescribe the function F as the following

$$F = c\left(\phi - \phi_{\text{target}}\right),$$

which allows a smooth transition between the initial ϕ to our target configuration, ϕ_{target}, with speed governed by the constant c. In our case, the target configuration is given by two cones whose zero level set is two identical circles of radius $x(s)$ (see Figure 4), i.e.,

$$\begin{aligned} \phi_{\text{target}} = \max&\left\{0, x(s) - \left|\boldsymbol{\xi} - \left(x(s),0\right)\right|\right\} \\ &+ \max\left\{0, x(s) - \left|\boldsymbol{\xi} + \left(x(s),0\right)\right|\right\}. \end{aligned} \tag{26}$$

Using this form of F and the given initial condition and terminal configuration, we wish to solve the following PDE for each crypt level, s,

$$\begin{cases} \dfrac{\partial \phi}{\partial \tau} = -c\left(\phi - \phi_{\text{target}}\right)|\nabla \phi| & \text{for } (\boldsymbol{\xi},\tau) \in \Omega \times (0,T] \quad (27) \\ \text{Periodic Boundary Conditions} & \text{for } \boldsymbol{\xi} \in \partial\Omega,\ \tau \in (0,T] \quad (28) \\ \phi(\boldsymbol{\xi},0;s) = \max\left\{0, x(s) - |\boldsymbol{\xi}|\right\} & \text{for } \boldsymbol{\xi} \in \Omega, \quad (29) \end{cases}$$

where Ω is a square domain in (ξ,ζ) space that completely encompasses the initial and terminal circles. Here, we have implemented periodic boundary conditions as this is the usual condition needed to track the propagating interface. To numerically solve the Hamilton-Jacobi-type PDE given by Equations (27) – (29), we use an entropy-satisfying scheme produced by Engquist and Osher [50] that converges to the viscosity solution. Figure 4 shows a series of snapshots in time of the solution for ϕ to Equations (27) – (29) using a radius of $x(s) = 1$. Also shown is the level set corresponding to the ϕ function. In the first panel, we can see the initial ϕ, where we have only considered the positive part of the cone and the negative portion is set to zero. The corresponding level set is a circe in the (ξ,ζ) plane with circumference equal to 2π. As ϕ evolves to the two cone configuration (here, also, the negative portion is set to zero), we see the level set begin to expand and pinch into the middle until

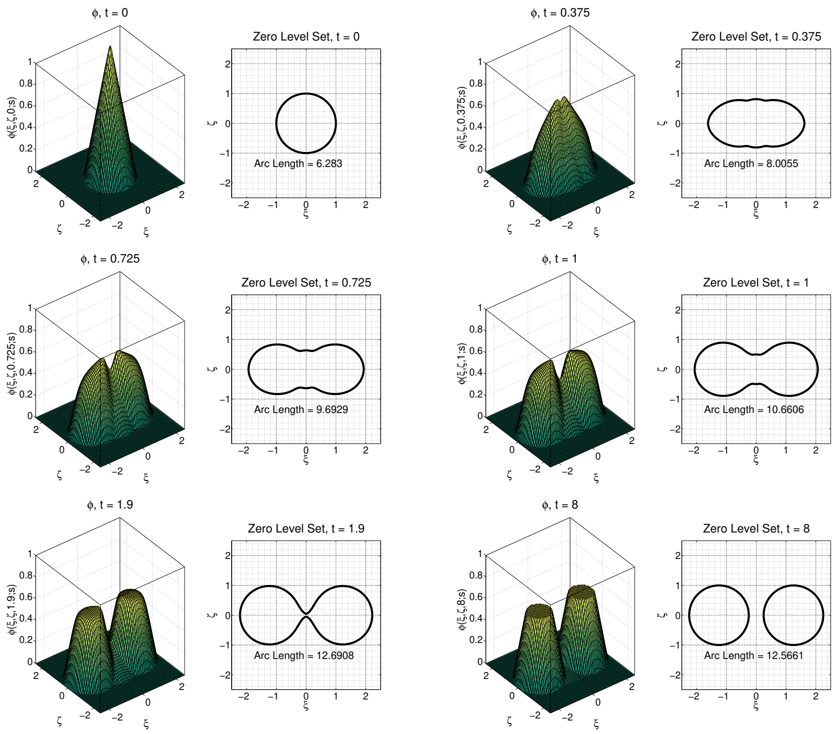

Fig. 4: A series of plots that show the evolution of ϕ as governed by Equations (27) – (29) for $x(s) = 1$. Next to each snapshot is the level set in (ξ, ζ) space. Parameters: $x(s) = 1$ and $c = 2$.

two complete circles of radius $x(s) = 1$ emerge. Furthermore, at steady-state, the arc-length of both circles totals 4π, exactly double that of the initial circumference. This method effectively captures geometrically the process of a single circle dividing into two identical circles.

To apply this to the full crypt domain, we implement the above method for each circle of radius $x(\hat{s})$ at every crypt level s. Furthermore, to simulate the cascade of dividing circles, we initiate the division of each crypt level after the level directly below it has experienced a 2% increase in arc-length. In this way, the crypt appears to divide as each crypt level takes its turn to split into two identical circles. Figure 5 shows the series of snapshots in time of the dividing crypt. We can see the initial crypt domain begins to bud at the base of the crypt, which is represented by the first circle of circumference one (representing a single cell) splitting into two circles each of circumference equal to one. Once this initial circle has an arc-length of 1.02 times its original arc-length, the next layer begins to split. This results in a cascade-type mechanism that eventually splits the entire crypt into two identical crypts, which can be seen in the third and fourth panels of the figure. Therefore, we have successfully applied the level set method to describe, geometrically, the crypt fissioning process.

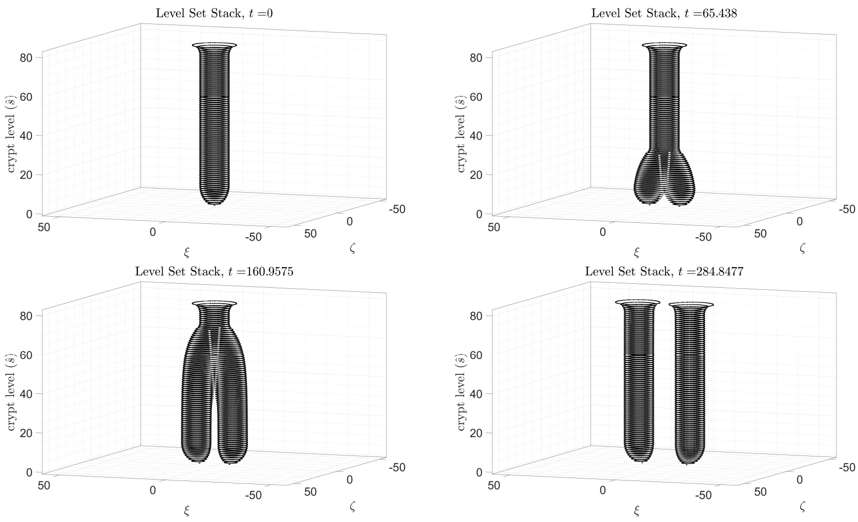

Fig. 5: A series of snapshots in time of the level set method applied to a stack of circles whose radii are prescribed by the parameterization $x(s)$ from Emerick et. al. [19].

4 PDE Coupled to Fission Process

We couple the one-dimensional Stefan problem considered in Section 2 to the crypt fissioning process discussed in the previous section. In this way, the cascade dividing mechanism that splits the crypt into two will be linked to the cell density at each level. The primary focus of this section is to simulate normal crypt fission and provide a simple model that works as a basis for further analysis and numerical computation.

We assume a crypt is already formed via Equations (19) – (24) with $q_1(s,\tau)$ representing the cell number density. A new crypt emerges due to overpopulation in the base. As before, we model the new crypt as a single chain of cells, given by the cell number density per unit arc-length, $q_2(s,\tau)$, with an advancing front, $L(\tau)$. However, in this model, we assume that the progression of the last cell is determined by the level set evolution of the crypt level in which that cell resides. In this sense, the length of the new crypt moves forward at the moment the current crypt level splits into two identical circles. Therefore, at the point of bifurcation, the force exerted on the cells from cellular proliferation is used to expand the crypt wall. To model this, we consider the following set of coupled equations,

$$
\begin{cases}
\dfrac{\partial q_2}{\partial \tau} = \dfrac{\partial}{\partial s}\left(\dfrac{\alpha}{q_2^2}\dfrac{\partial q_2}{\partial s}\right) & \\
\qquad + \left[\dfrac{\ln 2}{T_c} H(\sigma - \sigma_p) - mH(\sigma_s - \sigma)\right] q_2 & \text{for } s \in \big(0, L(\tau)\big), \tau > 0 \quad (30)\\
\dfrac{dL}{d\tau} = v_s(L,\tau)\left(1 + \kappa \left|\dfrac{\partial \phi}{\partial \tau}(\boldsymbol{\xi}, \tau; L)\right|\right)^{-1} & \text{for } \boldsymbol{\xi} \in \Omega, \tau > 0 \quad (31)\\
\dfrac{\partial \phi}{\partial \tau} = -v_s(L,\tau)F(s,\tau)\,|\nabla\phi| & \text{for } (\boldsymbol{\xi}, s) \in \Omega \times \big(0, L(\tau)\big), \tau > 0 \quad (32)\\
\dfrac{\partial q_2}{\partial s}(0,\tau) = 0 & \text{for } \tau > 0 \quad (33)\\
q_2(L,\tau) = 1 & \text{for } \tau > 0 \quad (34)\\
q_2(s,0) = 1 & \text{for } s \in \big(0, L(\tau)\big) \quad (35)\\
L(0) = 1 & (36)\\
\text{Periodic Boundary Conditions on } \phi & \text{for } (\boldsymbol{\xi}, s) \in \partial\Omega \times \big(0, L(\tau)\big), \tau > 0 \quad (37)\\
\phi(\boldsymbol{\xi}, 0; s) = \max\{0, x(s) - |\boldsymbol{\xi}|\} & \text{for } (\boldsymbol{\xi}, s) \in \Omega \times \big(0, L(\tau)\big), \quad (38)
\end{cases}
$$

where v_s is the velocity function, as before, given by

$$v_s(s,\tau) = -\frac{\alpha}{q_2(s,\tau)^3}\frac{\partial q_2}{\partial s}(s,\tau),$$

κ is a positive constant, and the function F is defined as

$$F(s,\tau) = (\phi - \phi_{\text{target}}) \begin{cases} H\big(q_1(0,\tau) - q_1^{\text{tol}}\big) & \text{for } s = 0 \\ H\big(L(\tau) - s\big) & \text{for } s > 0 \end{cases},$$

where ϕ_{target} is given by Equation (26) and $H(\cdot)$ is the Heaviside function. We note several key elements of the above system of equations: The evolution of the cell density that forms the new crypt is governed by the reduced equations studied in Section **??** and as the crypt length grows, we assume the new crypt is radially symmetric. The evolution of the first level set function (i.e. $\phi(\boldsymbol{\xi}, \tau; 0)$ is initiated when the cell population (measured by $q_1(s,\tau)$) at the base of the already formed crypt passes a certain threshold. This is evident from the function F above, which is switched on by the Heaviside function when $q_1(0,\tau) > q_1^{\text{tol}}$. Furthermore, the speed of this evolution is $v_s(L,\tau)$, which means the force required to push cells upward is redirected to the expansion of the crypt level. This expansion is measured by calculating the speed of the changing level set function, $|\partial\phi/\partial\tau|$, which is coupled to the rate of change of the new crypt length. The change in crypt length is equal to the last cell's velocity as in the Stefan formulation, but it is divided by the term

$$1 + \kappa \left|\frac{\partial \phi}{\partial \tau}(\boldsymbol{\xi}, \tau; L)\right|,$$

which measures the speed of the changing level set function at the current crypt level. It follows that $\partial\phi/\partial\tau > 0$ when each crypt level is dividing. However, as ϕ reaches the target steady-state, we have $\partial\phi/\partial\tau \to 0$. This means that the rate of change of the crypt length slows down when the current crypt level is dividing into two equal circles. Therefore, as cells divide, the spring force directs the cells upward until they reach a new crypt level. Once this crypt level is attained, the movement upward is antagonized, and cells move to form two new crypt levels before moving upward again. Lastly, the function F ensures that this progression continues because once the crypt length reaches a higher, undivided crypt level, the term $H\big(L(\tau)-s\big)$ initiates the evolution of the next level set function.

The model above describes a similar cascade effect as in the previous section, but it is linked to the cell density and cell velocity at each crypt level. Since the boundary condition at $s=L$ restricts the cell number density to be one at the moving boundary, the total number of cells at the bifurcating crypt level is equal to the arc-length of that crypt level. Once the arc-length doubles, the crypt length moves forward at the prescribed cell velocity and the chain of cells follows the radially symmetric cell motion and proliferation model. We can produce Figure 6, which shows a simulation of Equations (30) – (38) in dimensional units. Here, we can see that the progression of the cell number density is similar to that of the one-dimensional model, except the rate of change of the crypt length is slowed by the change in each crypt level. As the the crypt length progresses, we assume the established part is radially symmetric and the new crypt is essentially the interior parts of the two crypts.

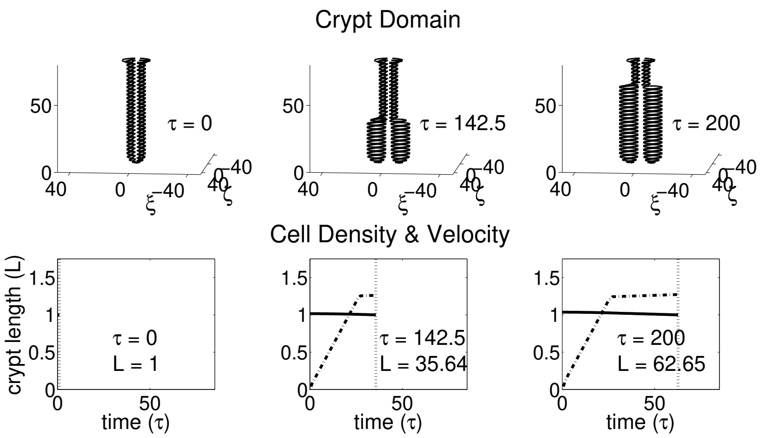

Fig. 6: Simulation of Equations (30) – (38). The top three panels show the progression of the dividing crypt for $\tau=0$, $\tau=119$, and $\tau=175$ hours. The bottom three panels show the progression of the cell density (solid), cell velocity (dashed-dotted), and crypt length (dotted). All parameters are the same as Figure 3 with $\kappa=1$.

Figure 7 shows a time-course plot of the crypt length with several values of the weight parameter κ. Here, we can see that when the length reaches a new crypt level, the rate of growth is reduced since the cell velocity is used to expand the crypt wall. This results in a step-like profile. For larger κ, this step-like plot is exaggerated in the sense that it takes longer for each crypt level to divide. If κ is small, this process will be smoothed out and the cascade of dividing crypt levels will occur more quickly.

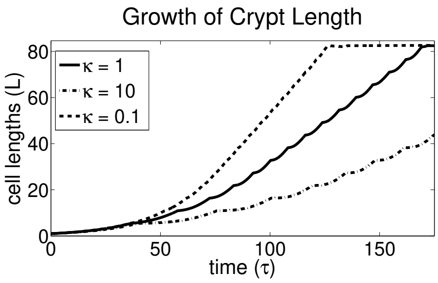

Fig. 7: A time-course plot of the numerical solution for $L(\tau)$ to Equation (31) for various values of κ. All simulations use $N_s = 16$ crypt levels.

This model provides a simple coupling mechanism between the one-dimensional cell population model and the geometric description of normal crypt fission as modeled by the level set method.

5 Discussion

The goal of this paper was to investigate the phenomenon of normal colonic crypt fission. The methods presented here expand on the previous models discussed in Emerick et. al. [19,7] and provide a mechanism by which a crypt forms and grows in one-dimensional space. We also consider the geometric progression of crypt fission in three-dimensional space. Using level set methods as presented by Enquist and Osher [48,49], we were able to capture the crypt bifurcation process and couple the evolution of the crypt with the cell velocity.

It is not clear how this model can be used to describe abnormal crypt fission in the presence of *APC* mutations. The target ϕ function completely determines the steady-state structure of the newly formed crypts; hence, any perturbations from the underlying APC regulation model must have an effect on the function F, which is a forcing function for the crypt level expansion. In this way, the crypt fissioning process may develop a branch instead of a fully sized crypt. For example, if the radius of one of the cones in ϕ_{target} is perturbed, the crypt will certainly divide into a normal-sized crypt and a smaller crypt. Further analysis on this topic will be conducted in the future.

We note here that further consideration is needed to establish the switch between cell movement up the crypt wall and cell expansion along the circumference of each crypt level. To this end, a two-dimensional representation of the full crypt may be required. That is, if we map the surface of revolution that represents the crypt to a two-dimensional plane, say in (s_1, s_2) space, further care can be taken in describing the movement in both directions. In this sense, the growth of the crypt length in the s_1-direction is described by the Stefan condition and the growth in the s_2-direction can be linked to the level set method. Moreover, the evolution of the cell number density can be modeled by the generalization of Leeuwen et. al.'s [16] model to twodimensions as presented by Murray et. al. [51]. Overall, this chapter is a first look into a more detailed model of tissue level dynamics in the human colon.

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